Vatsal Verma

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PROFESSIONAL SUMMARY

Backend Software Engineer with nearly 3 years of experience developing reliable and scalable applications using Java, Spring Boot, Apache Kafka, and cloud-native technologies. Experienced in building RESTful APIs, designing microservices, and delivering features for large-scale enterprise platforms serving millions of users. Hands-on experience in Generative AI and AI-powered automation, including intelligent solutions for incident triage and operational efficiency. Skilled in Docker, Kubernetes, CI/CD, and application security, with a strong focus on performance, code quality, and end-to-end feature ownership.

TECHNICAL SKILLS

Languages: Java, Python, SQL

Backend & Distributed Systems: Spring Boot, Spring Cloud, Spring Security, microservices architecture, REST APIs, OpenAPI, OAuth2/JWT, system design, object-oriented programming, data structures and algorithms

Data, Search & Messaging: Apache Kafka, Couchbase, Elasticsearch, event-driven architecture

AI / GenAI: Retrieval-Augmented Generation (RAG), RAG Ingestion and Retrieval Design, LLM-powered AI agents, GenAI, semantic search, vector database, MCP (Model Context Protocol), LangGraph, Google Agent Development Kit, Natural Language Processing, TensorFlow, Keras

DevOps & Testing: Docker, Kubernetes, Containerization, Red Hat OpenShift, Jenkins, Maven, Git, CI/CD, JUnit, TestNG

PROFESSIONAL EXPERIENCE

Software Developer Experienced *(SWE-2)* — *August 2023 – Present*
Amdocs, Pune, India

- Contributed to the development and delivery of business-critical features within a large-scale billing ecosystem comprising 36 microservices, supporting end-to-end feature lifecycles through Agile/Scrum methodologies for Tier-1 telecom clients including AT&T and Bell Canada, collectively serving 50+ million subscribers.
- Engineered scalable event-driven microservices and RESTful APIs using Java, Spring Boot, Spring Cloud, Apache Kafka, and Couchbase, while incorporating OpenAPI specifications, OAuth2/JWT-based authentication mechanisms, and API rate limiting to strengthen application security and performance.
- Architected a production-grade Agentic AI-powered automated incident triage solution with LangGraph and Google ADK, leveraging a complete RAG pipeline with hybrid retrieval (BM25 and dense vector search) and Elasticsearch (kNN, dense_vector) as the knowledge repository for embeddings and historical root cause analyses.
- Integrated real-time Apache Kafka event streams, REST APIs, and MCP (Model Context Protocol) tool interfaces to enrich AI agent context, enabling engineering teams to rapidly identify, diagnose, and reprocess exception records, ultimately reducing root cause analysis time by 20% and significantly decreasing manual incident handling efforts.
- Streamlined Maven-based CI/CD pipelines deployed on Red Hat OpenShift, improving build efficiency and reducing integration testing execution time by 25%.
- Managed secure production deployments and release cycles, ensuring high code quality and release readiness. Hardened production security by remediating 42 CVE and OWASP vulnerabilities.

AI/ML Research Intern — *August 2021 – February 2022*
Indian Institute of Technology (IIT-BHU), Varanasi, India

- Built facial recognition models using Convolutional Neural Networks (CNNs), HOG (Histogram of Oriented Gradients), TensorFlow, and Keras to support identification and classification of 50,000+ individuals.
- Applied computer vision techniques to agricultural research by analyzing unique muzzle patterns of over 100 cattle, extending facial recognition methodologies to livestock identification use cases.
- Investigated deep learning algorithms and image processing pipelines while effectively managing research timelines and presenting technical findings, analyses, and recommendations to faculty advisors.

PROJECTS

AI-Powered Animal Detection and Alert System — *Python, TensorFlow, Keras, OpenCV, REST API*

- Developed an end-to-end deep learning inference pipeline utilizing Convolutional Neural Networks (CNNs), TensorFlow, Keras, and OpenCV to enable real-time animal detection with an accuracy rate of 92.68%.
- Implemented image preprocessing and data augmentation techniques to improve model robustness and ensure reliable performance across varying environmental and lighting conditions.
- Integrated low-latency event notification capabilities using the Twilio REST API to deliver real-time alerts, earning recognition at WildHacks hackathon for innovative and creative API utilization.

NLP Document Summarization Service — *Java, Spring Boot, Spring REST, Maven, NLP*

- Constructed a Spring Boot-based REST microservice incorporating an NLP pipeline consisting of tokenization, lemmatization, sentence segmentation, and LexRank algorithm, achieving 85%+ summarization accuracy when validated against reference summaries.
- Established a modular ingestion and ranking architecture exposed through HTTP APIs by implementing standard Java microservice design patterns, including layered controllers, service components, and REST-based data transfer objects (DTOs).

EDUCATION

Vellore Institute of Technology, Vellore — *July 2019 – May 2023*
Bachelor of Technology in Computer Science and Engineering with specialization in IoT
CGPA: 8.22/10

ACHIEVEMENTS AND LEADERSHIP

- **Earned the Innovation Award at Amdocs** for enhancing TestNG suite execution efficiency and optimizing CI/CD processes through Kubernetes API-based cloud cluster integration.
- **Secured recognition at WildHacks 2021** for delivering the most creative implementation of the Twilio API during a competitive hackathon.
- **Led the design team for the ISA-VIT Student Chapter** by managing a team of 15 members and overseeing the design, development, and successful release of an application on the Google Play Store.